

SKILL WEEK 12

S23: Question1

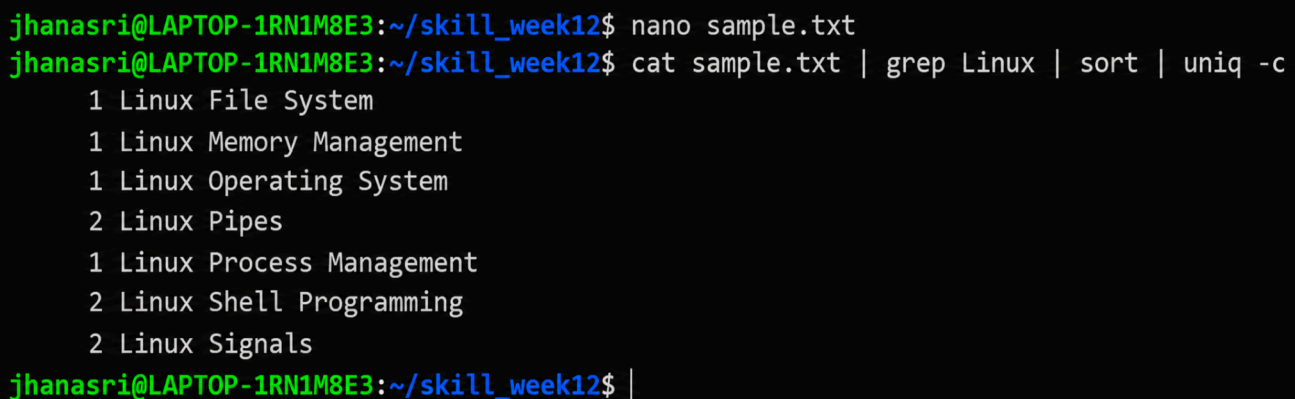
Aim:

To execute Linux pipelines, launch multiple background jobs, monitor running processes and system resources, detect job failures, and analyze performance using Linux commands.

A screenshot of a terminal window showing the nano 7.2 text editor editing a file named sample.txt. The file contains a list of Linux topics: Linux Operating System, Linux Shell Programming, Linux Process Management, Linux Signals, Linux Pipes, Linux Memory Management, Linux File System, Linux Shell Programming, Linux Signals, and Linux Pipes. The cursor is at the end of the last line.

```
GNU nano 7.2 sample.txt *
Linux Operating System
Linux Shell Programming
Linux Process Management
Linux Signals
Linux Pipes
Linux Memory Management
Linux File System
Linux Shell Programming
Linux Signals
Linux Pipes|
```

Output:

A screenshot of a terminal window showing the execution of a pipeline command. The user runs 'cat sample.txt | grep Linux | sort | uniq -c', which outputs a list of topics with their frequency: 1 Linux File System, 1 Linux Memory Management, 1 Linux Operating System, 2 Linux Pipes, 1 Linux Process Management, 2 Linux Shell Programming, and 2 Linux Signals.

```
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ nano sample.txt
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ cat sample.txt | grep Linux | sort | uniq -c
 1 Linux File System
 1 Linux Memory Management
 1 Linux Operating System
 2 Linux Pipes
 1 Linux Process Management
 2 Linux Shell Programming
 2 Linux Signals
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ |
```

```
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ sleep 200 &  
[1] 864
```

```
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ sleep 250 &  
[2] 899
```

```
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ sleep 300 &  
[3] 925
```

```
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ jobs -l  
[1]      864 Running                sleep 200 &  
[2]-    899 Running                sleep 250 &  
[3]+    925 Running                sleep 300 &
```

```
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ ps -f  
UID          PID     PPID  C  STIME TTY          TIME CMD  
jhanasri    293       292  0  05:52 pts/0        00:00:00 -bash  
jhanasri    864       293  0  05:53 pts/0        00:00:00 sleep 200  
jhanasri    899       293  0  05:53 pts/0        00:00:00 sleep 250  
jhanasri    925       293  0  05:54 pts/0        00:00:00 sleep 300  
jhanasri   1107       293  0  05:54 pts/0        00:00:00 ps -f
```

```
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ top  
top - 05:54:55 up 26 min,  1 user,  load average: 0.01, 0.01, 0.00  
Tasks:  24 total,  1 running, 23 sleeping,  0 stopped,  0 zombie  
%Cpu(s):  0.1 us,  0.2 sy,  0.0 ni, 99.7 id,  0.0 wa,  0.0 hi,  0.1 si,  0.0 st  
MiB Mem :  7782.6 total,  7237.8 free,   515.6 used,   190.1 buff/cache  
MiB Swap: 2848.0 total,  2848.0 free,    0.0 used.  7267.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21532	12960	9632	S	0.7	0.2	0:01.05	systemd
177	message+	20	0	9604	5372	4656	S	0.3	0.1	0:00.15	dbus-daemon
184	root	20	0	17968	8776	7752	S	0.3	0.1	0:00.07	systemd-lo+
1154	jhanasri	20	0	9308	5688	3492	R	0.3	0.1	0:00.02	top
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.00	init-system
6	root	20	0	1196	2132	2088	S	0.0	0.0	0:00.03	init
52	root	19	-1	58324	15888	14724	S	0.0	0.2	0:00.37	systemd-jo+
181	root	20	0	25484	6828	5156	S	0.0	0.1	0:00.11	systemd-ud+
111	systemd+	20	0	21468	13264	11084	S	0.0	0.2	0:00.07	systemd-re+
130	systemd+	20	0	91028	7960	6996	S	0.0	0.1	0:00.05	systemd-ti+
176	root	20	0	2444	2676	2424	S	0.0	0.0	0:00.00	cron
200	syslog	20	0	222516	5620	4552	S	0.0	0.1	0:00.00	rsyslogd
207	root	20	0	3124	1076	1836	S	0.0	0.0	0:00.00	agetty
208	root	20	0	107020	23192	13732	S	0.0	0.5	0:00.08	unattended+
291	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	sessionle+
292	root	20	0	3280	1252	1108	S	0.0	0.0	0:00.01	Relay(293)
293	jhanasri	20	0	6880	5440	3780	S	0.0	0.1	0:00.04	bash
294	root	20	0	6780	4668	3884	S	0.0	0.1	0:00.08	login
339	jhanasri	20	0	29312	11616	9496	S	0.0	0.1	0:00.07	systemd
340	jhanasri	20	0	21160	3600	1872	S	0.0	0.0	0:00.05	(sd-pam)
362	jhanasri	20	0	6880	5356	3664	S	0.0	0.1	0:00.01	bash
864	jhanasri	20	0	3132	1908	1792	S	0.0	0.0	0:00.00	sleep
899	jhanasri	20	0	3132	1908	1792	S	0.0	0.0	0:00.00	sleep
925	jhanasri	20	0	3132	1896	1784	S	0.0	0.0	0:00.00	sleep

```
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ kill %2
```

```
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ jobs -l  
[1]      864 Running                sleep 200 &  
[2]-    899 Terminated            sleep 250 &  
[3]+    925 Running                sleep 300 &
```

```
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ time ls -R
```

```
..:
```

```
report.txt  skill12_task1  skill12_task2  test_cases.sh  
sample.txt  skill12_task1.c skill12_task2.c
```

```
real    0m0.004s  
user    0m0.003s  
sys     0m0.000s
```

```
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ |
```

Observation:

1. Linux pipelines processed data through multiple commands.
2. Multiple background jobs were executed simultaneously.
3. Running processes were monitored using `ps -f`.
4. System resource usage was monitored using `top`.
5. A background job was terminated using `kill`.
6. Command execution time was measured using `time`.

Result:

Successfully executed Linux pipelines, launched multiple background jobs, monitored running processes and system resources, detected job termination, and analyzed command performance using Linux commands.

S23: Question2

Aim:

To document the architecture of a Linux program, explain its execution path and system calls, create a flow diagram, record design decisions, and review the program documentation.

Program:

```
GNU nano 7.2 skill12_task2.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Program Started\n");

    printf("Process ID : %d\n", getpid());
    printf("Parent Process ID : %d\n", getppid());

    printf("Executing Program...\n");

    sleep(2);

    printf("Program Finished Successfully\n");

    return 0;
}
```

Output:

```
sys      0m0.000s
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ nano skill12_task2.c
[1]+  Done                  sleep 200
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ gcc skill12_task2.c -o skill12_task2
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ ./skill12_task2
Program Started
Process ID : 2310
Parent Process ID : 293
Executing Program...
Program Finished Successfully
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ |
```

```

jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ cat skill12_task2.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Program Started\n");

    printf("Process ID : %d\n", getpid());
    printf("Parent Process ID : %d\n", getppid());

    printf("Executing Program...\n");

    sleep(2);

    printf("Program Finished Successfully\n");

    return 0;
}
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ nan special
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ nan spiepid
[1]+  Done                  sleep 300
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ strace ./skill12_task2
Command 'strace' not found, but can be installed with:
sudo apt install strace
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ sudo apt update
[sudo] password for jhanasri:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
248 packages can be upgraded. Run 'apt list --upgradable' to see them.
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ sudo apt install strace
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following new package will be installed:
  strace
0 upgraded, 1 newly installed, 0 to remove and 248 not upgraded.
Need to get 584 kB of archives.
After this operation, 1311 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 strace amd64 6.8-1ubuntu2 [584 kB]
Fetched 584 kB in 3s (233 kB/s)
Selecting previously unselected package strace.
(Reading database ... 94825 files and directories currently installed.)
Preparing to unpack .../strace_6.8-1ubuntu2_amd64.deb ...
Unpacking strace (6.8-1ubuntu2) ...
Setting up strace (6.8-1ubuntu2) ...
Processing triggers for man-db (2.12.0-4build2) ...
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ strae ./skill12_task2
execve("./skill12_task2", ["/skill12_task2"], 0x7ffc2be8030 /* 27 vars */) = 0
brk(NULL)                               = 0x559509f4a000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fad0e45a000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=197211, ...}) = 0
mmap(NULL, 197211, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7fad0e152000
close(3)                                = 0

```

```

jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ ./skill12_task2
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f9b390ef000
arch_prctl(ARCH_SET_FS, 0x7f9b390ef700) = 0
set_tid_address(0x7f9b390efad0) = 3357
set_robust_list(0x7f9b390efae0, 24) = 0
rseq(0x7f9b390f0f60, 0x20, 0, 0x53053053) = 0
mprotect(0x7f9b390ef000, 8192, PROT_READ) = 0
mprotect(0x7f9b390f0000, 4096, PROT_READ) = 0
mprotect(0x7f9b390f1000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7f9b390f1000, 19712) = 0
stat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
getrandom("\xa7\x94\x1b\x6b\x8a\x01\x0b\x78", 8, GRND_NONBLOCK) = 8
brk(NULL) = 0x559509f4a000
brk(0x559509f4b000) = 0x559509f4b000
write(1, "Program Started\n", 16Program Started
) = 16
getpid() = 3357
write(1, "Process ID : 3357\n", 18Process ID : 3357
) = 18
getppid() = 3346
write(1, "Parent Process ID : 3346\n", 25Parent Process ID : 3346
) = 25
write(1, "Executing Program...\n", 21Executing Program...
clock_nanosleep(CLOCK_REALTIME, 0, {tv_sec=2, tv_nsec=0}, 0x7ffc2a237030) = 0
write(1, "Program Finished Successfully\n", 30Program Finished Successfully
) = 30
exit_group(0)
jhanasri@LAPTOP-1RN1M8E3:~/skill_week12$ |

```

Observation:

1. getpid() returned the current process ID.
2. getppid() returned the parent process ID.
3. strace displayed the sequence of system calls made by the program.
- 4.

The program executed successfully and terminated using exit_group().

Result:

Successfully documented the program architecture, explained the execution path, identified system calls using strace, and verified the program execution flow in Linux.

S24: Question1

Aim:

To organize source code into separate folders, create a build script, add a README file, verify the repository structure, and validate the project deliverables in Linux.

Program:

```
GNU nano 7.2 src/main.c *
#include <stdio.h>

int main()
{
    printf("OSSP Skill Week 12 Project Build Successful!\n");
    return 0;
}
```

```
GNU nano 7.2 include/calculator.h *
#ifndef CALCULATOR_H
#define CALCULATOR_H

void displayMessage();

#endif
```

```
GNU nano 7.2 docs/README.md *
# OSSP Skill Week 12

## Project Structure
- src : Source code
- include : Header files
- docs : Project documentation

## Build Command
gcc src/main.c -o project_build

## Run Command
./project_build
```

```
GNU nano 7.2 build.sh *
#!/bin/bash

echo "Building OSSP Project..."

gcc src/main.c -o project_build

echo "Build Completed Successfully."
```

Output:

```
sindhuja@SINDHUJA: ~/skill_ x + v
sindhuja@SINDHUJA:~$ cd skill_week12
sindhuja@SINDHUJA:~/skill_week12$ mkdir src include docs
sindhuja@SINDHUJA:~/skill_week12$ touch src/main.c
sindhuja@SINDHUJA:~/skill_week12$ touch include/calculator.h
sindhuja@SINDHUJA:~/skill_week12$ touch docs/README.md
sindhuja@SINDHUJA:~/skill_week12$ ls -R
.:
docs    report.txt  skill12_task1  skill12_task2  src
include sample.txt  skill12_task1.c skill12_task2.c test_cases.sh

./docs:
README.md

./include:
calculator.h

./src:
main.c
sindhuja@SINDHUJA:~/skill_week12$ nano src/main.c
sindhuja@SINDHUJA:~/skill_week12$ nano include/calculator.h
sindhuja@SINDHUJA:~/skill_week12$ nano docs/README.md
sindhuja@SINDHUJA:~/skill_week12$ nano build.sh
sindhuja@SINDHUJA:~/skill_week12$ chmod +x build.sh
sindhuja@SINDHUJA:~/skill_week12$ ./build.sh
Building OSSP Project...
Build Completed Successfully.
sindhuja@SINDHUJA:~/skill_week12$ ./project_build
OSSP Skill Week 12 Project Build Successful!
sindhuja@SINDHUJA:~/skill_week12$ tree
.
├── build.sh
├── docs
│   └── README.md
├── include
│   └── calculator.h
├── project_build
├── report.txt
├── sample.txt
├── skill12_task1
├── skill12_task1.c
├── skill12_task2
├── skill12_task2.c
├── src
│   └── main.c
└── test_cases.sh

4 directories, 12 files
sindhuja@SINDHUJA:~/skill_week12$ |
```

Observation:

1. The project was organized into separate folders.
2. A README file documented the project.
3. A build script automated compilation.
4. The project compiled and executed successfully.
5. The repository structure was verified using the tree command.

Result:

Successfully organized the source code, created a build script, added a README file, verified the repository structure, and validated the project deliverables in Linux.

S24: Question2

Aim:

To create demo scenarios, validate Linux features, demonstrate pipelines and signals, verify program execution, and perform a final project review in Linux.

```
GNU nano 7.2 demo.txt *
Linux Operating System
Linux Shell
Linux Pipes
Linux Signals
Linux Processes
Linux Memory
Linux Shell
Linux Signals
Linux Pipes|
```

Output:

Observation:

1. A background process was created using the sleep command.
2. The process was suspended using the SIGSTOP signal.
3. The suspended process was resumed using the bg command. 4.
- The process was terminated successfully using the kill command. 5.
- The project directory structure was verified using ls -R.

Result:

Successfully created demo scenarios, validated Linux process management, demonstrated background job control using signals, and completed the final project review in Linux.